Vibrations Post-Lab: MATLAB Script Session

Conversation transcript with Claude Code · 28 October 2026 · repository sblanco12/SBCode, branch claude/beautiful-bardeen-1ermz5

This document reproduces the user's messages and Claude's replies. Tool activity (file inspection, test runs) is summarised in italics rather than shown in full. The figures are Octave test renders of the script, made inside the session; in MATLAB the legends are placed automatically and look cleaner.

1. Request: MATLAB script for the post-lab

User

Attached: Vibrations_datasets.zip, Vibrations_Simulation.zip

I'm completing a post-lab worksheet for a vibrations lab. The experiment uses a cantilever beam at three lengths (16 in, 20 in, 24 in), tested two ways (a shaker test and an impact hammer test), plus a simulation. I need MATLAB code that processes my data and generates every required plot. Please write one well-commented MATLAB script, with a clearly labeled section or function for each part. It should run in standard MATLAB with no extra toolboxes. Flag every assumption you make, both in the code comments and in your reply.

I'm attaching 2 zip files: (1) Vibrations_Simulation.zip: the simulation output files; (2) Vibrations_datasets.zip: the physical shaker and hammer experiment data.

Data selection

- The datasets zip contains a folder called "Vibrations datasets" with one subfolder per lab section (M002, M003, ... M009) plus an Instructions.txt file.
- Use ONLY the files inside the "Vibrations datasets/M008" subfolder. Do not use data from any other section's folder.
- Within M008, use ONLY group B's data. Shaker test: only the "_GROUP B" files ("16 IN _GROUP B.txt", "20 IN _GROUP B.txt", "24 IN _GROUP B.txt"); ignore the "_GROUP A" files. Hammer test: only the "in_group b" files (e.g. "10_9_2026m008_hammer_16 in_group bfft.txt" and "...bdat.txt"); ignore the "in_group a" files.
- The script should locate these folders reliably (e.g. let me set the folder paths at the top, and/or search for them or ask me to pick the folder if they aren't found).

File formats

- Shaker file: the last 1-2 rows may be repeated or irrelevant and can be removed.
- Hammer files (.txt, tab-delimited, one header line): per Instructions.txt, "dat" files are the frequency plot data and "fft" files are the time-series data.
- Simulation files (.txt, one header line, 2 columns): "...fft.txt" holds the frequency response magnitude vs frequency, and "...dat.txt" holds the phase angle vs frequency. Please confirm this and the phase units from the data.

Simulation file names

- Effect of length (3.1): 3.1_16in, 3.1_20in, 3.1_24in
- Effect of damping (3.2), 20 in beam, carbon steel, no end mass: 3.2_20in (c = 1), 3.2_20in_3 (c = 3), 3.2_20in_5 (c = 5), 3.2_20in_7 (c = 7), 3.2_20in_9 (c = 9 kg/s)
- Effect of end mass (3.3), 20 in beam, carbon steel, c = 0.01: 3.3_20in_0.25, 3.3_20in_0.40, 3.3_20in_0.65 (kg)
- Effect of material (3.4), 20 in beam, c = 0.01, end mass = 0: 3.4_20in (carbon steel), 3.4_20in_SS (stainless steel), 3.4_20in_Al (aluminum)
- The actual file names have a prefix like "10_9_2026group27" and a suffix of "fft.txt" or "dat.txt", and the material tags may be lowercase.

Simulation conditions

- Length 20 in (508 mm) for 3.2, 3.3 and 3.4. Width and thickness match section 3.1.
- 3.2: c = 1, 3, 5, 7, 9 kg/s, end mass 0, carbon steel
- 3.3: end mass = 0.25, 0.40, 0.65 kg, c = 0.01, carbon steel
- 3.4: carbon steel, stainless steel, aluminum, c = 0.01, end mass 0

Plots only, no tables.

Shaker experiment (7 plots, M008 group B data)

- 3 plots, one per beam length: Amplitude vs. Frequency of the output, with uncertainty (error bars, horizontal and vertical). Each plot must include a theoretical response curve. Because the measured voltage has an unknown gain, scale the theoretical curve by (a) multiplying the theoretical response R by ω^2, then (b) multiplying by the ratio of the measured maximum response to the maximum of that modified theoretical response.
- 3 plots, one per length: the same as above, but Amplitude vs. Normalized Frequency, where the x-axis is divided by the theoretical undamped natural frequency. Include uncertainties.
- 1 plot: Undamped natural frequency (fn) and resonance frequency (fr) vs. beam length, with error bars. Use the damping ratio from the hammer test (group b). Label the legend as fr and fn.

Hammer experiment (6 figures, M008 group b data)

- 3 figures (one per length), each with 4 subplots: voltage vs. time (input), voltage vs. time (output), amplitude vs. frequency (input, via FFT), amplitude vs. frequency (output, via FFT). The frequency plots need the x-axis labeled "Frequency (Hz)" and the y-axis labeled "Amplitude."
- 3 figures (one per length), each with 2 subplots: magnitude of the frequency response |H(f)| vs. frequency, and phase lag vs. frequency. The y-axes should be labeled "|H(f)|" and "Phase lag."

Simulation (4 figures, 2 subplots each)

For each parameter study below, make one figure with two subplots that overlay the parameter values (one line per parameter value, with a legend): (top) magnitude of the response |H(f)| vs. frequency, (bottom) phase lag vs. frequency. The y-axis labels should be "|H(f)|" and "Phase lag."

- Figure A, length (3.1): 16, 20, 24 in
- Figure B, damping coefficient (3.2): c = 1, 3, 5, 7, 9 kg/s, loaded from the 3.2_20in files
- Figure C, end mass (3.3): 0.25, 0.40, 0.65 kg, loaded from the 3.3_20in_... files
- Figure D, material (3.4): carbon steel, stainless steel, aluminum, loaded from 3.4_20in, 3.4_20in_SS, 3.4_20in_Al

Total: 17 figures (7 shaker + 6 hammer + 4 simulation).

Ask for clarification if confused

Work done in this turn: extracted both zips; inspected the M008 group B shaker and hammer files and all 28 simulation files; identified the hammer channels and the dat-file scaling; inferred the beam size from the simulation; prototyped the damping estimate; installed Octave and ran the script end to end (all 17 figures rendered); committed and pushed vibrations_postlab.m.

Claude

I wrote the script **vibrations_postlab.m** and pushed it to claude/beautiful-bardeen-1ermz5. It makes all 17 figures in your order: shaker 1–7, hammer 8–13, simulation 14–17. It needs no toolboxes and expects MATLAB R2019b or newer.

I couldn't test it in MATLAB itself. I ran it in Octave: every figure rendered and the numbers below came out of that run. Run it once in MATLAB before you submit.

Finding the folders: you can set m008Folder and simFolder at the top. If you leave them empty, the script searches the current folder and its own folder. If that fails, it unzips any datasets or simulation zip it finds there into a temp folder. As a last step it asks you to pick the folder. It only reads files in M008, and of those only the _GROUP B shaker files and the in_group b hammer files.

Your format questions:

- **Simulation files:** confirmed. fft is |H|, with no units and equal to 1 at 0 Hz. dat is the phase angle in **degrees**: it runs from 0 to about −180. The script plots phase lag as −phase, so 0 to 180°.
- **Hammer channels:** column 3 is the hammer (input) and column 2 is the beam (output). The script detects this from the shape of the signal, since the hammer is a short spike. The resonance peaks only show up in column 2.
- **Hammer dat files:** they are the same spectrum as an FFT of the time data, divided by √2. The script computes its own FFT; set hammerUseDatFile = true to plot the dat values instead.

Results:

L	fn theory	fr (shaker)	ζ (hammer)
16 in	31.81 Hz	27 ± 0.5 Hz	0.0017
20 in	20.36 Hz	17 ± 0.5 Hz	0.0023
24 in	14.14 Hz	13 ± 0.5 Hz	0.0043 ± 0.0008

Assumptions (each is tagged [A#] in the code):

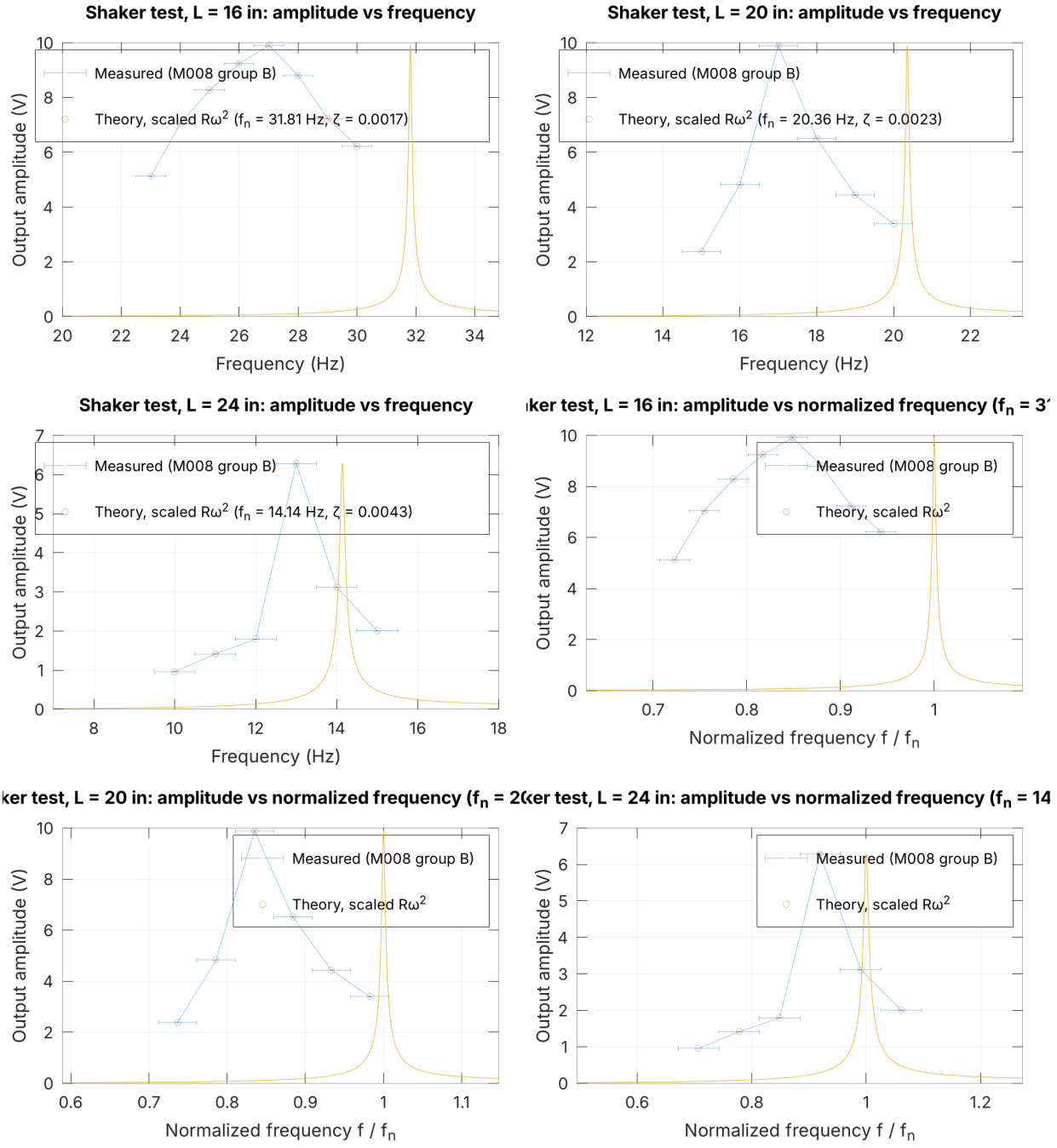
- **Beam size (please check this one first): 1.00 in wide × 0.25 in thick.** Your message didn't give the dimensions, so I worked them out from the simulation. A beam of this size reproduces the simulated natural frequencies to within 0.06%. If your handout says otherwise, change beamWidth_in and beamThick_in at the top.
- **Carbon steel:** E = 200 GPa, ρ = 7850 kg/m^3.
- **Theoretical fn model:** k = 3EI/L^3 and effective mass = 33/140 of the beam mass. This is the model the simulation matches. No accelerometer or end mass is included; you can add one with endMass_kg.
- **Hammer input channel:** picked automatically as described above. You can force it with hammerInputCol.
- **Damping ratio:** I filtered the output to keep only the first mode and fitted the slope of its decay. The uncertainty is the spread of fits over three parts of the decay. I didn't use the half-power method because the FFT resolution (0.37 Hz) is wider than the resonance peak.
- **H(f):** the plain output/input voltage ratio from the single hit, with no averaging and no sensor calibration.
- **Shaker cleaning:** when a frequency appears more than once, only the first reading is kept. This drops the last 2 rows of each file. In the 24 in file those rows (0.023 and 0.047 V) are clearly junk. The script prints every row it removes.
- **Shaker uncertainty:** frequency ±0.5 Hz, half the 1 Hz sweep step; voltage: half the 0.001 V display resolution, combined with a 0.4% repeatability taken from the repeated readings. You can change these at the top.
- **fr and fn:** fr is the frequency of the largest shaker reading. fn = fr/√(1−2ζ^2). Because ζ is so small, **fn and fr overlap in Fig 7**; that's expected.
- **Length uncertainty in Fig 7:** ±1/32 in.

Things in your data worth discussing in the worksheet:

- **Measured peaks sit 8–15% below theory.** So in Figs 1–6 the sharp theoretical peak lands to the right of your data, and your data peaks at f/fn ≈ 0.85–0.92. Added mass at the attachment or a flexible clamp would explain this.
- **The hammer phase plot drifts steadily with frequency.** The output's largest peak comes about 20 ms after the hit, which suggests a delay between the two channels. I left the phase as measured.
- **Possible double hit at 16 in.** The script warns about a second hammer spike at 31% of the main one. Per your instructions, it still uses the M008 data.
- **The simulation phase is stuck at exactly −90° for several points near resonance.** That comes from the simulation output itself, so I used the simulated |H| peak, not the phase, to check the beam size.

Set saveFigures = true to also save every figure as a PNG.

Figures from the first version (Octave test render)



Figs 1–6: shaker amplitude vs frequency and vs normalized frequency (before the axis edit in turn 2).

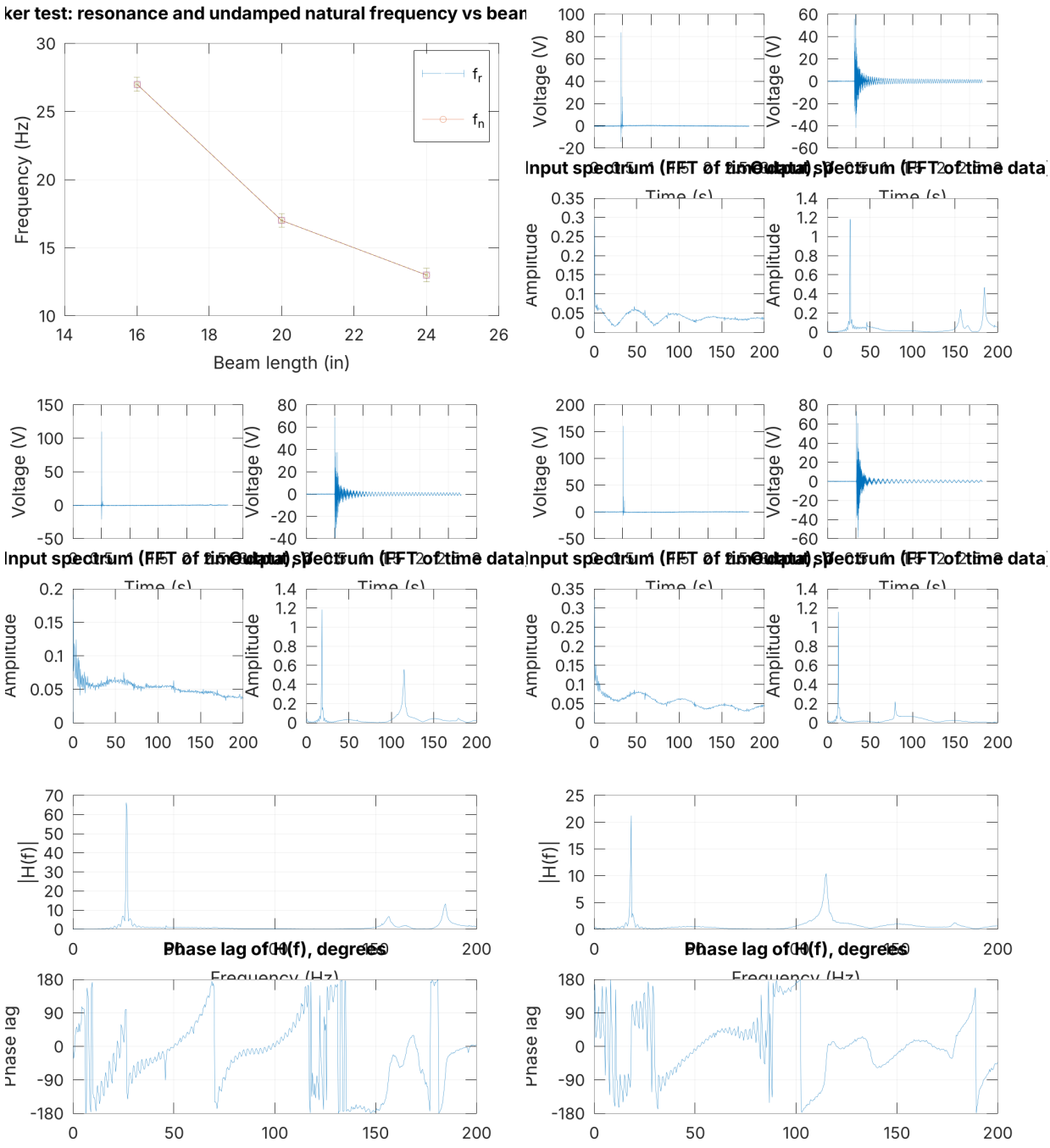


Fig 7: fr and fn vs length; Figs 8–12: hammer time/FFT plots and |H(f)| with phase lag.

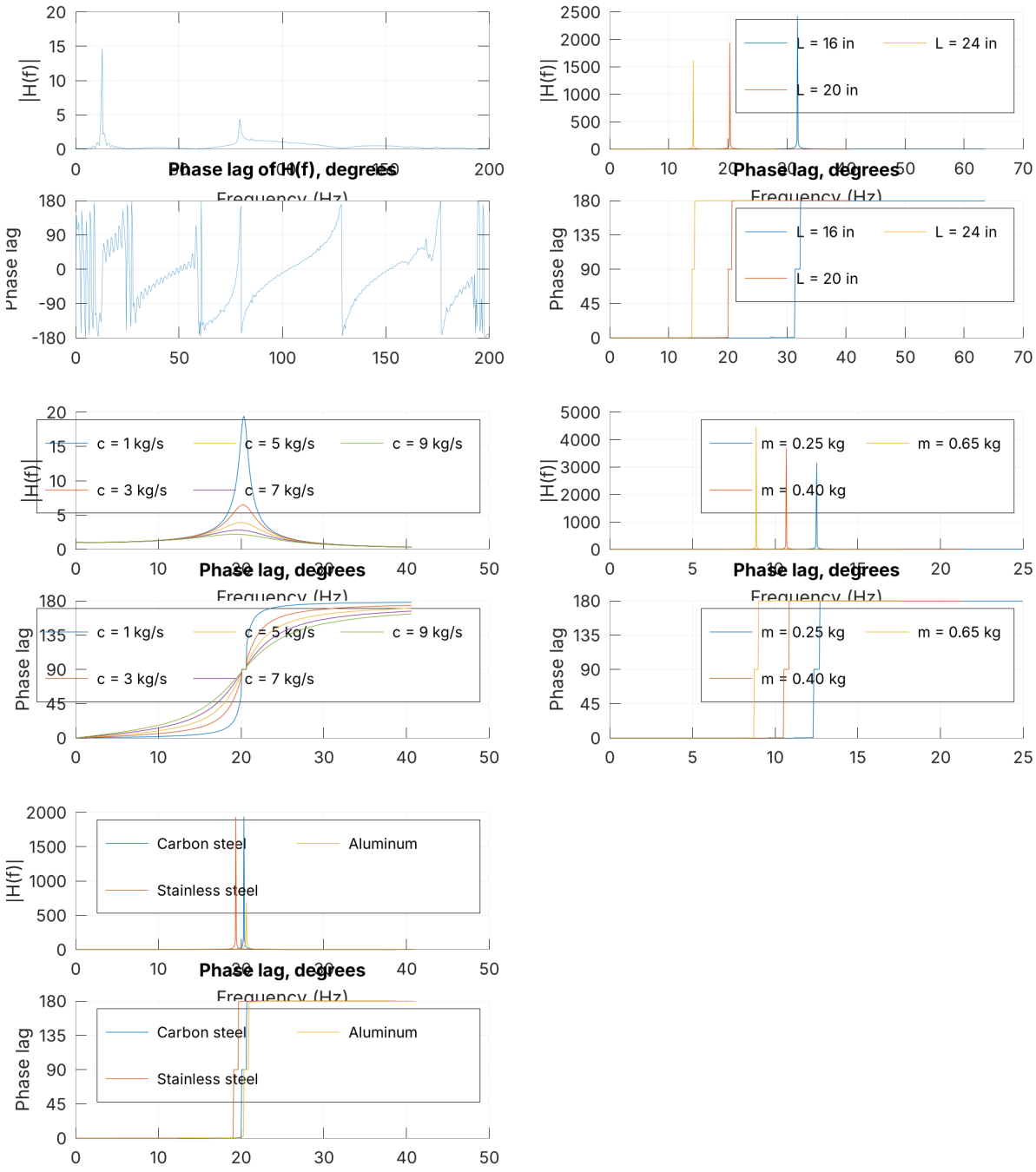


Fig 13: hammer |H(f)| and phase lag for 24 in; Figs 14–17: simulation studies 3.1–3.4.

2. Request: zoom out the plots

User

edit the code to zoom out on the plots, y axis count by intervals of 2 instead of 1 and maybe try centering it a little more for x axis. tell me which line of code was edited pls

Work done in this turn: applied the change to the shaker amplitude plots (Figs 1–6), re-rendered them in Octave to check, then committed and pushed.

Claude

I zoomed out the shaker amplitude plots (Figs 1–6), set the y-axis to count by 2 V, and centred the x-axis better. I applied this to the shaker plots only, since those were the ones counting by 1. If you meant other figures, tell me which. I re-rendered the six plots in Octave to check them, and pushed the change to the same branch.

Lines edited in vibrations_postlab.m:

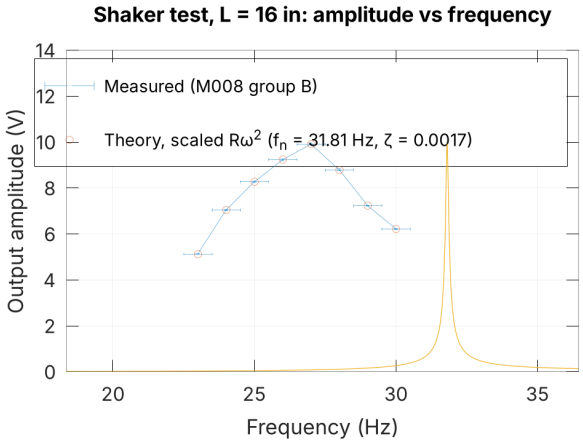
Lines	What changed
116–118 (new, in USER SETTINGS)	Three new settings: xZoom = 1.6 (how far the x-axis zooms out), yHeadroom = 1.3 (top of the y-axis is 1.3× the largest reading) and yTickStep = 2 (y-axis counts by 2).
274–280 (replaced the old 2-line fLo/fHi calculation)	The x-window is now centred halfway between your data and the theoretical peak, then widened by xZoom. Line 280 sets the y-axis top, rounded up to a multiple of 2.
290 (new, Figs 1–3)	ylim([0 yTop]); set(gca, 'YTick', 0:yTickStep:yTop);
303 (new, Figs 4–6)	The same line for the normalized-frequency plots.

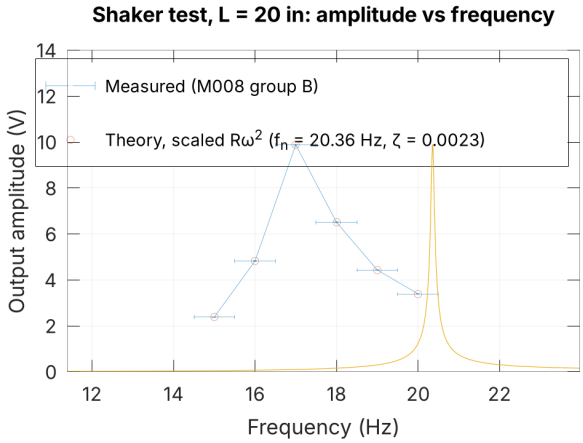
Result:

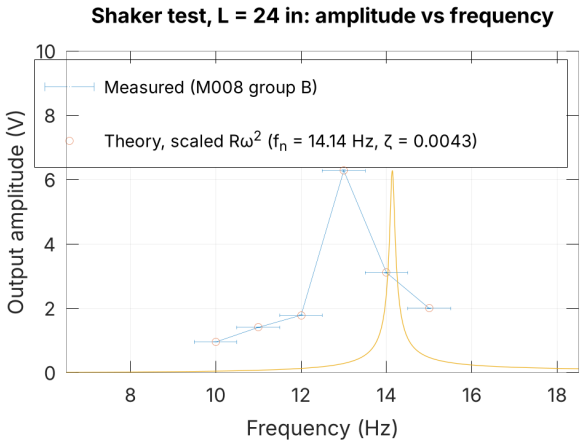
- **16 in and 20 in:** the y-axis now runs 0–14 V in steps of 2, and the extra room clears space for the legend.
- **24 in:** it runs 0–10 V.
- **x-axis:** both the data and the theory peak now sit near the middle of each plot, with margin on both sides.

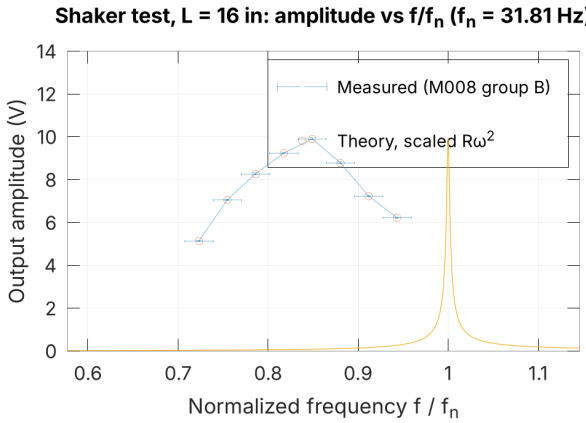
To adjust further, change the settings on lines 116–118. A larger xZoom zooms out more horizontally, and a larger yHeadroom adds more space at the top.

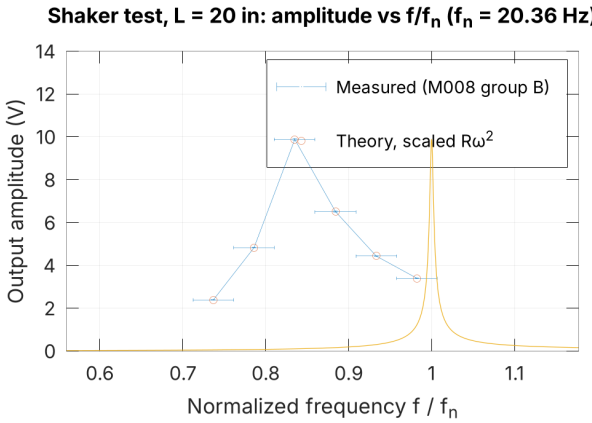
Figs 1–6 after the edit (Octave test render)

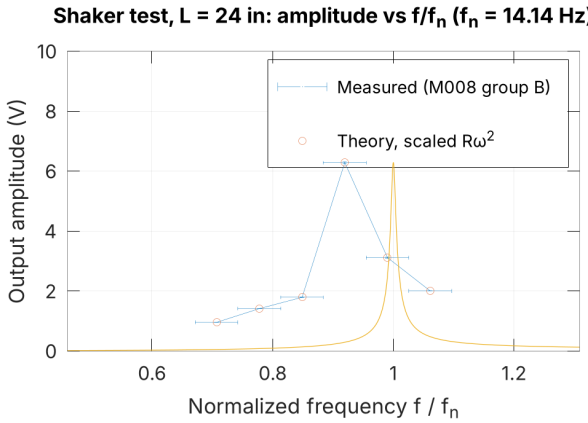












3. Request: PDF of the conversation

User

Make a pdf document of this conversation

Claude

This document.